

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43

COURSE OUTCOME COVERED

CO -2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Design Task

Weightage: 20%

QUESTIONS:

Total Marks: 20

Question 1: Responsive Navigation Bar Design - Marks(4)

Suppose that an e-commerce website requires a responsive navigation bar that adapts across desktop, tablet, and mobile devices.

- (a) Identify key components required in a navigation bar for an e-commerce website.
 - (b) Design a responsive navigation bar using appropriate CSS techniques.
 - (c) Demonstrate how the navigation adapts to smaller screens (e.g., hamburger menu).
 - (d) Evaluate the usability and suggest improvements for better user experience.
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Question 2: Product Hover Effect - Marks(4)

Suppose that an e-commerce website wants to enhance user interaction through visual effects on product items.

- (a) Identify suitable CSS properties for hover effects.
 - (b) Suggest appropriate **CSS animation or transition** for product interaction.
 - (c) Design a hover effect for a product card.
 - (d) Evaluate performance and user experience considerations.
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Question 3: Layout Selection (Flexbox vs Grid) - Marks(4)

Suppose that a dashboard layout needs to be designed for a web application.

- (a) Identify layout requirements of a dashboard (rows, columns, alignment).
 - (b) Compare **Flexbox and Grid** based on layout needs.
 - (c) Choose the most suitable layout technique and justify your choice.
 - (d) Evaluate the scalability and responsiveness of the chosen approach.
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Question 4: Mobile-First Blog Layout - Marks(4)

Suppose that a blog page must be designed following a mobile-first approach.

- (a) Identify key sections of a blog layout.
 - (b) Design a mobile-first layout using CSS.
 - (c) Demonstrate how the layout scales for larger screens using media queries.
 - (d) Evaluate accessibility and readability of the design.
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Question 5: JavaScript Event Handling for Dynamic Menu - Marks(4)

Suppose that a website includes a dynamic menu that responds to user interactions.

- (a) Identify appropriate JavaScript events for menu interaction.
- (b) Design event handling logic for menu toggle functionality.
- (c) Implement interaction for user actions (click/hover).
- (d) Evaluate performance and suggest improvements.

Code of Conduct:

I S.Ganesh Reddy /192311153 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

S.Ganesh Reddy

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Understanding of Concept	25	Demonstrates complete understanding of design	Good understanding with minor gaps	Basic understanding with some errors	Poor understanding or incorrect concepts

		requirements and concepts			
Design / Implementation	30	Well-structured, efficient, and responsive design implementation	Correct implementation with minor issues	Basic implementation with inconsistencies	Incorrect or incomplete implementation
Use of Tools (CSS/JS)	20	Effective and advanced use of tools/techniques	Appropriate use of tools	Limited or inconsistent use	Incorrect or minimal use
Analysis & Evaluation	15	Insightful analysis with strong justification and optimization	Good analysis with justification	Basic analysis with limited depth	Poor or no analysis
Documentation / Clarity	10	Clear, well-structured, and properly presented solution	Mostly clear with minor issues	Basic clarity, missing details	Poor or unclear documentation

ANSWERS

Question 1: Responsive Navigation Bar Design (4 Marks)

- (a) A responsive e-commerce navigation bar should include a company logo, navigation links (Home, Products, Categories, About, Contact), search bar, shopping cart icon, user account/login option, and a hamburger menu for mobile devices.
 - (b) A responsive navigation bar can be designed using CSS Flexbox to align items horizontally on larger screens. Flexible widths, relative units, and media queries help the layout adapt to different screen sizes.
 - (c) On smaller screens, the navigation links are hidden and replaced with a hamburger menu. When the menu is clicked, the navigation links are displayed vertically, making them easier to access on mobile devices.
 - (d) The navigation bar should have clear labels, touch-friendly buttons, fast loading, and smooth animations. Adding sticky navigation, highlighting the active page, and ensuring keyboard accessibility further improve the user experience.
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Question 2: Product Hover Effect (4 Marks)

- (a) Suitable CSS properties for hover effects include transform, transition, box-shadow, opacity, background-color, color, and border.
 - (b) CSS transitions provide smooth effects such as scaling, rotating, changing colors, and adding shadows when the user hovers over a product card.
 - (c) A product card can slightly enlarge using the scale() transform, display a shadow, and change the button color when the mouse pointer moves over it. This attracts user attention and improves interaction.
 - (d) Hover effects should be lightweight to maintain good performance. Excessive animations should be avoided, and effects should remain smooth and responsive across all devices.
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Question 3: Layout Selection (Flexbox vs Grid) (4 Marks)

- (a) A dashboard typically contains a header, sidebar, main content area, widgets, charts, tables, and footer. The layout requires multiple rows, columns, proper alignment, and spacing.
 - (b) Flexbox is best suited for one-dimensional layouts (either rows or columns), whereas CSS Grid is designed for two-dimensional layouts involving both rows and columns. Grid provides better control over complex dashboard structures.
 - (c) CSS Grid is the most suitable choice because dashboards require multiple sections arranged in rows and columns. It offers better flexibility, alignment, and organization than Flexbox for complex layouts.
 - (d) CSS Grid is highly scalable and responsive. Combined with media queries, it allows the dashboard to automatically rearrange sections for tablets and mobile devices while maintaining a clean layout.
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Question 4: Mobile-First Blog Layout (4 Marks)

- (a) A blog layout typically includes a header, navigation menu, featured image, blog content, sidebar, comments section, and footer.

- (b) In a mobile-first approach, the layout is initially designed for small screens using a single-column structure. Content is stacked vertically with readable font sizes and responsive images.
 - (c) Media queries are used to adjust the layout for larger screens. On tablets and desktops, the sidebar appears beside the main content, additional spacing is added, and multiple columns may be introduced.
 - (d) The design should use readable fonts, sufficient spacing, proper color contrast, responsive images, keyboard accessibility, and semantic HTML to improve accessibility and readability.
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Question 5: JavaScript Event Handling for Dynamic Menu (4 Marks)

- (a) Common JavaScript events for menu interaction include click, mouseover, mouseout, mouseenter, mouseleave, and keydown for keyboard navigation.
- (b) The event handling logic detects when the user clicks the hamburger menu. JavaScript then toggles the visibility of the navigation links by adding or removing a CSS class.
- (c) When the user clicks or hovers over the menu, JavaScript displays the navigation items. Clicking again hides the menu, providing a simple and interactive navigation experience.
- (d) Performance can be improved by minimizing DOM manipulation, using event delegation where appropriate, reducing unnecessary event listeners, and ensuring keyboard accessibility for users with different input methods.